

TADDLEBOX — Feature Document

Version 1.0 · React Native (Expo) · Platform: iOS & Android

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1. Product Overview

TADDLEBOX is a gamified social platform built for college students. It combines a social media feed, community spaces, competitive mini-games, and an XP-based reward economy into a single app. Users earn XP by posting content, attending events, winning games, and maintaining daily login streaks — all of which can be converted into real cash rewards via a built-in wallet.

Core Value Pillars

Pillar	Description
Social	Post, comment, like, share and follow other students
Compete	Play games (Chess, Ludo, Block Blaster, Candy Connect) for XP
Earn	Accumulate XP through activity; convert to real cash
Discover	Browse events, hackathons, webinars, and communities
Grow	Level up, collect badges, climb leaderboards

2. Authentication & Onboarding

2.1 Splash Screen

- Animated app logo with bounce + glow fade sequence
- Auto-transitions to Onboarding after 2.5 seconds
- Full-screen dark gradient background

2.2 Onboarding Carousel

- 3-slide FlatList carousel with snap scrolling
- Each slide has: gradient background, large emoji illustration, title, subtitle
- Topics: Social Feed, Earn XP, Play & Win
- Skip button (top-right) and Next/Get Started buttons

2.3 Welcome Screen

- Feature highlights: XP rewards, games, communities, events
- Two CTAs: **Login** and **Register**

2.4 Registration (3-Step Flow)

Step 1 — Account Details

- Full name, email, password, confirm password
- Real-time validation with inline error messages
- Progress indicator (1/3)

Step 2 — Profile Info

- Username / handle (with @ prefix)
- College name
- Bio (optional)
- Progress indicator (2/3)

Step 3 — Interests

- Grid of interest chips (e.g. Tech, Gaming, Design, Startup, Music, Sports)
- Multi-select toggle — tap to add/remove
- Progress indicator (3/3)
- Submit triggers OTP verification

2.5 OTP Verification

- 6-digit code input with individual boxes
- Auto-focus advances cursor to next box
- Auto-submits when all 6 digits are filled
- Resend button with 30-second countdown timer
- Edit phone number link

2.6 Login

- Email + password form
- Inline validation errors
- Loading state (spinner in button) during submission
- Forgot Password link → ForgotPasswordScreen

2.7 Forgot Password

- Email input field
- Submit shows confirmation UI: success icon, instructions, Back to Login button

3. Home Feed

The Home tab is the app's central experience, combining a social feed with gamification widgets.

3.1 Header Bar

- Hamburger menu (opens Side Drawer)
- Centered logo "TADDLEBOX"
- Notification bell with unread count badge

3.2 Mini Status Cards

Two cards displayed side-by-side below the header:

Card	Color	Taps to
 Streak (N Days)	Amber	Streak detail modal
 Total XP	Purple	Wallet screen

The XP card bounces with a spring animation when a Daily Reward is claimed (XP fly-to-card effect).

3.3 Stories Row

- Horizontal scrollable list of story avatars
- First item is always the user's own story (+ icon)
- Unseen stories: animated gradient ring border
- Seen stories: grey ring border
- Tap any avatar → StoryViewer screen

3.4 Spotlight Carousel

- Horizontal snap-scroll carousel of featured items
- 3-4 cards: hackathons, competitions, platform announcements
- Each card shows: gradient background, emoji, title, subtitle, tag label, metadata
- Dot indicator for position

3.5 Trending Chips

- Horizontal scrollable filter bar: All , #Hackathon , #GameTime , #CollegeFest , #DevLife , #StudyTips
- Tap a chip to filter the post feed
- Active chip: filled purple background + bold text

3.6 Daily Login Reward Card

- Shown once per day (auto-dismissed after claim)
- Shows: gift icon, title "Daily Login Reward", progress track, Claim button
- On claim:
 1. Gift icon bounces (spring animation)
 2. "+50 XP" text floats upward and fades
 3. Golden "⚡ +50 XP" pill particle flies from card to XP mini-card
 4. XP mini-card springs to 1.4× scale then snaps back
 5. Card fades + slides down and removes itself after 1.8 s

3.7 Post Feed

- Vertical list of PostCard components
- Filtered by active trending chip
- Pull-to-refresh (900ms delay, themed tint color)

- Empty state message if no posts match active filter

3.8 Side Drawer

- Slides in from left with backdrop overlay
- Header: user avatar, name, handle
- Navigation links: Notifications, Bookmarks, Leaderboard, Settings, Wallet, Events
- Badge counts on relevant items
- Tap outside or back button closes drawer (280ms transition delay to prevent flicker)

3.9 Streak Modal (Sheet)

- Appears from bottom when Streak card is tapped
- Shows 7-day M-T-W-T-F-S-S grid with completion status per day
- Today highlighted with fire emoji
- "Next reward in N more days" section
- Full milestone list: 3d (+50 XP), 7d (Exclusive Badge), 14d (₹5 Cash), 30d (Elite Status)
- Tap backdrop or close button to dismiss

4. Social Features

4.1 Post Card

Each post card shows:

- Author avatar, name, handle, post timestamp
- Community tag (colored pill) + XP earned pill
- Post text content
- Optional media: emoji banner OR actual image (from media picker), with aspect ratio support (1:1 or 16:9)
- Hashtag display
- Action bar: ❤️ Like (with count), 💬 Comment (with count), 🔗 Share, 🌟 Bookmark

Interactions:

- Like: Toggles liked state, count updates optimistically
- Bookmark: Toggles saved state, saved posts appear in Bookmarks screen
- Comment: Navigates to CommentsScreen with post context
- Share: Native OS share sheet with post text
- Author tap: Navigates to UserProfileScreen

4.2 Create Post (FAB)

Accessible via the center FAB (⊕) in the bottom tab bar.

CreatePostModal fields:

- Post content text input (multi-line, character count)
- Hashtag input: type and add chips; tap chip to remove
- Media picker: photo or video from device library; choose aspect ratio
- Community selector: dropdown to post to a specific community
- Submit button (disabled when empty)

4.3 Comments Screen

- Post preview card at the top

- Threaded comment list (FlatList)
- Each comment: avatar, name, handle, text, timestamp, like count
- Nested replies support
- Text input at bottom for composing a new reply
- Keyboard-aware layout (avoids keyboard overlap)

4.4 Bookmarks Screen

- Header with total saved count badge
 - Flat list of saved PostCards (fully interactive)
 - Empty state: 🚫 icon + "No bookmarks yet" message
 - Works identically to the main feed for interactions
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5. Stories

5.1 Story Row Component

- Scrollable avatar list embedded in Home feed
- Your story (first) shows a ⊕ icon; tap to create
- Other stories: colored ring = unseen, grey = seen

5.2 Story Viewer Screen

- Full-screen immersive viewer
 - Segmented progress bar at top (1 segment per story, 5s each)
 - Tap left half of screen: go to previous story
 - Tap right half: go to next story
 - Press and hold: pause progress
 - Release: resume
 - Each story: gradient background, emoji, text content, username
 - Swipe down or back button exits
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6. Communities

6.1 Community Browse (CommunityScreen)

- Category filter tabs: All, Joined, Tech, Gaming, Lifestyle, Startup, Creative, Study
- Community cards in a scrollable grid/list:
 - Banner emoji + gradient background
 - Community name, description snippet
 - Member count
 - Join/Leave toggle button
 - Private badge for private communities
- Leaderboard section showing top XP earners in communities

6.2 Community Detail (CommunityDetailScreen)

- Banner header (gradient per category + emoji)
- Community avatar (gradient circle with emoji)
- Join/Joined button (gradient vs outline)
- Stats row: Members, Posts, Category
- Write post bar (only visible if joined)
- Feed filter tabs: All, 🔥 Trending, ✨ New

- Filtered post feed (posts belonging to or tagged with this community)
- Create post FAB → CreatePostModal pre-filled with community

Categories & their colors:

Category	Gradient
Tech	Deep purple → dark blue
Lifestyle	Forest green → teal
Gaming	Burnt orange → red
Startup	Olive → amber
Creative	Violet → pink
Study	Navy → cyan

7. Events

7.1 Event Feed (EventsScreen)

- Filter chips: All,  Live,  Online,  Offline,  Contest
- Featured event at top (larger card with prominent CTA)
- Event card list showing:
 - Event type icon + gradient
 - Title, type label (color-coded badge)
 - Date, time, location
 - XP reward chip + cash prize chip (if applicable)
 - Registration count + Register/Registered button

7.2 Event Types & Colors

Type	Emoji	Color
Hackathon		Purple
Webinar		Blue
Meetup		Green
Competition		Amber
Workshop		Teal

7.3 Registration Flow

- Tap Register → button switches to "✓ Registered" state
 - Registration count increments locally
 - XP reward displayed prominently as incentive
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8. Games

8.1 Game Hub (GamesScreen)

Three-tab layout:

- 1. **All Games** — browse and launch games
- 2. **Tournaments** — join live competitive events
- 3. **History** — past match results

Player Stats Bar (on All Games tab):

- Total XP earned from games
- Games played count
- Win rate percentage
- Current win streak

8.2 Available Games

Game	Emoji	Max XP
Chess	♟️	150 XP
Ludo	🎲	100 XP
Block Blaster	💣	80 XP
Candy Connect	🎯	60 XP

Each game card shows: gradient thumbnail, emoji, name, players online count, max XP, "HOT" badge if trending.

8.3 Play Modes

Mode	Icon	Description	XP
vs Bot	🤖	Practice vs AI, no entry fee	Full
Quick Match	⚡	vs random online player	Full
Tournament	🏆	Entry: 50 XP, compete for prizes	Bonus

8.4 Game Play Flow (GamePlayModal)

Lobby Phase:

- Game hero card with gradient + emoji
- Info chips: Max XP, Avg. Game time, Status
- Mode selector (Bot / Quick / Tournament) with radio buttons
- Play Now button (gradient)

Countdown Phase:

- Full-screen overlay
- Game emoji + "You vs AI Bot / Opponent"
- 3 → 2 → 1 → GO! animated countdown (spring scale)

Playing Phase:

- Score box (current score + XP to earn)
- Game name + elapsed time
- Live game board (visual simulation):
 - **Chess:** 8×8 board with pieces, move highlights
 - **Ludo:** 4-quadrant colored board with token circles
 - **Block Blaster:** 5×8 colored cell grid with line-clear animation
 - **Candy Connect:** 5×5 emoji grid with swap simulation
- Move log (Chess/Ludo): scrolling list of moves with player vs AI indicator
- Auto-advances moves every 1.4 seconds
- Game ends after ~9.8 seconds

Result Phase:

- Win/Loss card (spring scale-in animation)
- Win (🏆): green gradient card | Loss (😞): red gradient card
- Stats: XP Earned, Score, Mode
- Buttons: Exit | Play Again (gradient)
- Win → triggers XP earn in WalletContext + adds match to GamesContext

8.5 Tournaments Tab

- Tournament cards showing: game, name, prize (₹), XP prize, progress bar (% filled), entry fee, ends-in timer
- Register button → Alert confirming entry fee deduction
- Registered state: checkmark badge

8.6 Match History Modal

- Filter chips: All | 🏆 Wins | 😞 Losses
- Match rows: game emoji, name, opponent, duration, played-at, result badge, XP earned, score
- Empty state per filter

8.7 Leaderboard Modal

- Top 3 podium (🥇 / 🥈 / 🥉) with emoji avatars + XP
- Full list for ranks 4+: rank number, avatar, username, XP

9. Wallet & Monetization

9.1 Balance Cards

Two stacked gradient cards:

- **Cash Card:** ₹ balance + "Total Earned" + "Total Withdrawn" stats
- **XP Card:** XP balance with conversion rate (100 XP = ₹1)

9.2 Earn More Section

Grid of 4 action shortcuts:

Action	XP Range	Navigates to
Win Games	+50–150 XP	Games tab
Post Content	+10–75 XP	Create post

Attend Events	+25–200 XP	Events tab
Daily Streak	+10–50 XP	Home tab

9.3 Transaction History

- Filter chips: All | Earned | Spent | XP | Cash
- Transaction rows: type icon, title, date, amount (color-coded: gold for XP, red for negative, green for positive)
- Transaction types: earn, spend, convert, withdraw

9.4 Withdraw Modal

- Quick-select amounts: ₹100, ₹250, ₹500 chips
- Custom amount text input
- Validation: minimum ₹100, must not exceed balance
- Linked UPI display + "Change" button
- Confirms with warning about processing time

9.5 Link UPI Modal

- UPI ID text input with format validation (user@provider)
- Quick-select buttons for major apps: Paytm, GPay, PhonePe, BHIM
- Verify button → 1.5s mock verification → ✓ Verified state
- Save & Link button

9.6 Convert XP → Cash Modal

- XP amount input
- Live calculation display: N XP = ₹M (at ₹0.01/XP)
- Validation: minimum 500 XP, must not exceed balance
- Predefined amounts: 500, 1000, 5000 XP chips
- Confirm button

9.7 Wallet Settings Modal

- **Security:** Enable PIN, Enable Biometrics
- **Linked Accounts:** UPI ID display + link/change, Bank account
- **Notifications:** XP credits, Withdrawal updates, Promo offers

10. Gamification System

10.1 XP (Experience Points)

Earn XP by:

Activity	XP Range
Creating a post	+10–75 XP
Winning a game	+50–150 XP
Daily login reward	+50 XP
Attending an event	+25–200 XP

Daily streak bonus	+10-50 XP
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XP is displayed in:

- XP mini-card on home screen
- XP progress bar on profile
- Leaderboard rankings
- Wallet XP balance

10.2 Levels & Ranks

Users progress through levels (1-∞) with named ranks:

Rank	Approximate Level
Newcomer	1-4
Explorer	5-9
Pioneer	10-14
Champion	15-19
Legend	20+

Level badge shown: profile avatar, level-up screen, leaderboard.

10.3 XP Progress Bar

- Shows current XP / XP needed for next level
- Gradient fill (purple → cyan)
- Glowing dot at fill position
- Level number circle badge (left)
- Rank label

10.4 Daily Streak

- Tracked per consecutive login day
- 7-day visual calendar grid (M-T-W-T-F-S-S)
- Today's day shown with 🔥, completed days with ✓

Milestone Rewards:

Days	Reward
3 days	+50 XP Bonus
7 days	Exclusive Badge 🌟
14 days	₹5 Cash Reward 💎
30 days	Elite Status 🏆

10.5 Badges

Achievement badges displayed on profile:

Badge Color	Significance
Gold	Top achievements (Streak Legend, etc.)
Purple	Community or skill badges
Cyan	Event participation badges
Green	Social or XP milestone badges
Locked (greyed)	Not yet unlocked

10.6 Leaderboard

- Global ranking by total XP
 - Top 3 shown in podium view (🏆 🥈 🥉) with avatar emojis
 - Full list shows rank number, avatar, username, XP
 - Game-specific leaderboards in the Games tab
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11. User Profiles

11.1 Own Profile (ProfileScreen)

- Top bar: handle + share icon + settings icon
- Hero section (purple gradient): avatar (gradient circle), name, handle, rank, bio
- Stats row: Posts, Followers, Following, Total XP
- Edit Profile button + QR code share
- XP Progress Bar component
- Achievements row (horizontal scroll of badge cards)
- Community stats: joined count, games played, college
- Tab switcher: Posts | Media | Saved | Games
- Post grid (3-column, emoji-based thumbnails with like count)

11.2 Other User Profile (UserProfileScreen)

Accessed by tapping any post author.

- Same layout as own profile
 - Follow / Following toggle button (+ count update)
 - Message button
 - More (⋮) options button
 - No edit access
-

12. Notifications

12.1 Notification Types

Type	Color	Description
like	Pink	Someone liked your post
comment	Blue	Comment on your post

follow	Purple	New follower
mention	Cyan	Someone @mentioned you
event	Amber	Event reminder or update
achievement	Gold	Badge earned or level up

12.2 Grouping

Notifications grouped into:

- **Today** — within last 24h
- **Yesterday** — prior day
- **Earlier** — all older

12.3 Actions

- Mark all as read (removes unread indicators)
- Badge count on notification bell icon in Home header
- Read/unread visual distinction (opacity)

13. Settings & Personalization

13.1 Profile Section

- Display name and avatar
- Edit profile shortcut
- Account type (Student / Creator)

13.2 Privacy Settings

Toggle	Default	Effect
Public Account	ON	Posts visible to all
Activity Status	ON	Show online/last seen
Allow Tagging	ON	Others can tag in posts
Leaderboard Visibility	ON	Appear in public rankings

13.3 Theme

- Dark / Light mode toggle
- Instantly applies across all screens via ThemeContext
- Persists for session duration

13.4 Security

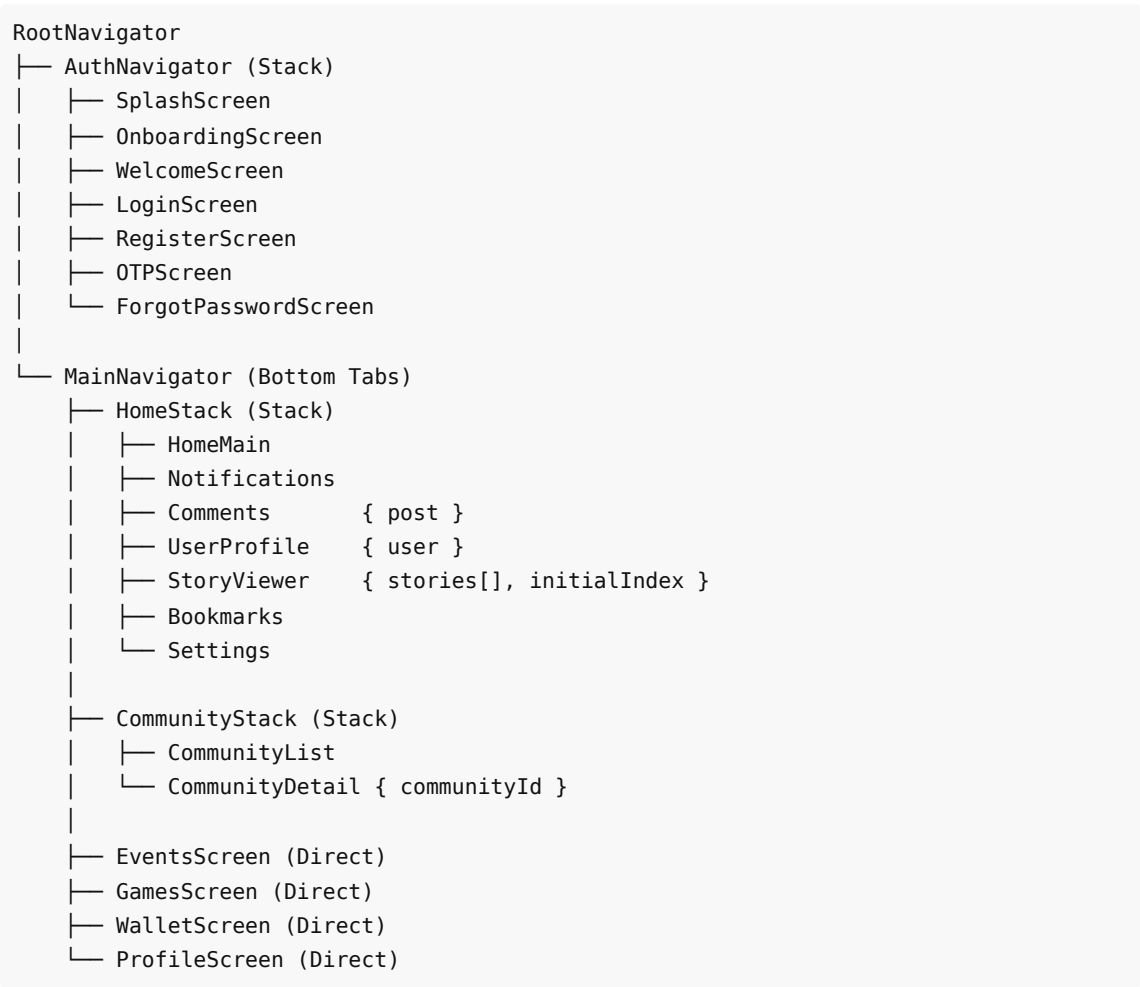
- Change Password
- Two-Factor Authentication (2FA) toggle
- Active Sessions list

13.5 Account Actions

- Logout (clears auth state)

- Delete Account (destructive, with confirmation)

14. Navigation Architecture



Bottom Tab Bar (6 tabs): Home | Community | Events | Games | Wallet | Profile

Center FAB (⊕) between Games and Wallet launches CreatePostModal.

15. Data & State Management

15.1 Context Providers

Context	State Managed	Actions
ThemeContext	isDark, colors	toggleTheme
AuthContext	isLoggedInIn	signIn, signOut
PostsContext	posts[]	addPost, toggleLike, toggleSave, deletePost
CommunityContext	communities[]	toggleJoin, addCommunity

WalletContext	balances, transactions[], settings	withdraw, convertXP, earnXP, linkUPI, toggleSetting
GamesContext	matches[], stats	addMatch

15.2 Wallet Currency System

- **XP:** Earned through activity; displayed everywhere gamification is present
- **INR:** Earned from XP conversion or event prizes; withdrawable via UPI
- **Exchange Rate:** 100 XP = ₹1 (minimum 500 XP to convert)
- **Minimum Withdrawal:** ₹100

15.3 Mock Data (Current)

All data is seeded in-memory with mock data. The architecture is ready for API integration by swapping context actions with API calls:

- Current user: Arjun Kumar, Level 12 Pioneer, 12,840 XP
- Initial cash: ₹1,250
- 3 initial posts, 4 communities, 6 game matches in history